SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL 3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3 Duration : 1 Hour

Total Marks:30

Annexure A

<u>Constraint-Based Problem Solving</u>

Code of Conduct:

I _G.Bhavya / 192411114 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

<u>Constraint-Based Problem Solving</u>

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that: Constraints:

   i. D1 cannot work Night shift ii. D2 must work before D3 (Morning < Afternoon < Night) iii. Only one doctor per shift iv. D3 cannot work Morning

   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

Problem Understanding

A hospital needs to assign **3 doctors (D1, D2, D3)** to **3 shifts (Morning, Afternoon, Night)** while satisfying all the given constraints.

Constraint Analysis

The constraints are:

- **D1 cannot work Night shift.**
- **D2 must work before D3** (Morning < Afternoon < Night).
- **Only one doctor can be assigned to each shift.**
- **D3 cannot work Morning shift.**
- **Each doctor must be assigned exactly one shift.**

Method Selection

Algorithm Used: Backtracking Search

Reason: Backtracking assigns one doctor at a time. If any constraint is violated, it backtracks and tries another assignment until a valid schedule is found.

Solution Development

Step 1: Assign **D1 = Morning** ✅
Constraint Check:

- D1 is not in Night ✓

Current Assignment:

Doctor	Shift
D1	Morning
D2	—
D3	—

Step 2: Assign **D2 = Afternoon** ✅
Constraint Check:

- Afternoon is after Morning.
- D2 must work before D3 ✔

Current Assignment:

Doctor	Shift
D1	Morning
D2	Afternoon
D3	—

Step 3: Assign **D3 = Night** ✅
Constraint Check:

- D3 is not in Morning ✔
- D2 (Afternoon) is before D3 (Night) ✔
- Only one doctor per shift ✔
- All doctors assigned exactly one shift ✔

Final Schedule

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

Backtracking Search Tree

```
Start
  |
D1 → Morning ✓
  |
D2 → Afternoon ✓
  |
D3 → Night ✓
  |
Goal Reached
```

Application of AI Concepts

- **Problem Type:** Constraint Satisfaction Problem (CSP)
- **Search Technique:** Backtracking Search
- **Variables:** D1, D2, D3
- **Domain:** Morning, Afternoon, Night
- **Constraints:** Given scheduling rules

Justification

Backtracking Search is suitable because it checks every assignment against the constraints. If a constraint is violated, it immediately backtracks and tries another assignment, ensuring a valid schedule is found efficiently.

Final Answer

- **Algorithm Used:** Backtracking Search
- **Valid Schedule:**
  - **D1 → Morning**
  - **D2 → Afternoon**
  - **D3 → Night**
- **All constraints are satisfied successfully.**

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1 iii. Cannot pass through obstacles iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

Note: Your question mentions a **5 × 5 grid with obstacles**, but the **grid diagram is missing**. Without the grid, it is **not possible** to determine the exact node expansion, optimal path, or total cost.

Below is the **exam-ready answer format**. Once you upload the grid, I can fill in the exact BFS traversal.

Problem Understanding

A robot must move from the **Start (S)** to the **Goal (G)** in a **5 × 5 grid**. The robot can move only **Up, Down, Left, and Right**. Obstacles cannot be crossed, and each move has a cost of **1**.

Constraint Analysis

The constraints are:

- Movement allowed only in **Up, Down, Left, and Right** directions.
- Cost of each move = **1**.
- Obstacles cannot be crossed.
- The robot must find the shortest path.
- Manhattan Distance can be used as the heuristic.

Method Selection

Algorithm Used: Breadth-First Search (BFS)

Reason:
Since every move has the **same cost (1)**, BFS always finds the **shortest path** from the start node to the goal node.

Solution Development

Step 1: Insert the **Start node (S)** into the queue.

Queue:

```
[S]
```

Step 2: Remove **S** from the queue and expand all valid neighbouring nodes.

Queue:

```
[Adjacent Nodes]
```

Step 3: Expand the first node in the queue and add all unvisited valid neighbours.

Queue:

```
[Next Level Nodes]
```

Step 4: Continue expanding nodes level by level until the **Goal (G)** is reached.

Node Expansion Table

Step	Expanded Node	Queue After Expansion
1	S	Adjacent Nodes
2	First Adjacent Node	Updated Queue
3	Next Node	Updated Queue
...	...	...
Final	G	Goal Reached

Cost Calculation

Each movement costs **1**.

Total Cost = Number of moves from S to G

Application of AI Concepts

- **Problem Type:** Path Finding
- **Search Algorithm:** Breadth-First Search (BFS)
- **State Space:** Grid cells
- **Goal Test:** Reach the Goal (G)

Justification

Breadth-First Search is the most suitable algorithm because all movements have equal cost. It explores nodes level by level and guarantees the shortest path from the start to the goal.

Final Answer

- **Algorithm Used:** Breadth-First Search (BFS)
- **Optimal Path:** Determined after exploring the grid.
- **Total Cost:** Equal to the number of moves from **S** to **G**.

Upload the 5 × 5 grid image for Question 2, and I'll solve it completely with:

- Step-by-step node expansion
- Queue updates after every step
- Final shortest path
- Total cost
- Exam-ready answer matching your rubric.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved. c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

c) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Problem Understanding

An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot starts from **S** and must reach the survivor at **G**. It operates in a **partially observable environment**, where some paths are blocked and new information is discovered while moving.

(a) Constraint Analysis

The given constraints and their impact are:

- **Move only Up, Down, Left, Right:** The robot can move only in four directions, limiting its possible actions.
- **Obstacles cannot be crossed:** The robot must find an alternative path whenever an obstacle is encountered.
- **Limited sensing capability:** The robot can observe only adjacent cells, so it must update its knowledge while exploring.
- **Movement cost = 1:** Every movement increases the total path cost.
- **Risky zones have an additional cost of 2:** The robot should avoid risky areas whenever possible to minimize the total cost.
- **Online search:** The robot must continuously update its decisions as it receives new information about the environment.

(b) Solution Approach

Search Strategy: Uniform Cost Search (UCS) with an Online Search Agent

Steps:

1. Start from the initial position **S**.
2. Observe the adjacent cells.
3. Ignore blocked paths and identify all possible moves.

4. Calculate the total cost of each possible move.
5. Expand the path with the lowest cumulative cost.
6. Update the map whenever new obstacles or risky zones are detected.
7. Continue the search until the robot reaches the survivor at **G**.

(c) Application of AI Concepts

Intelligent Agent

- The robot senses its surroundings, processes the information, and takes suitable actions to reach the survivor.

Rationality

- The robot always selects the action that minimizes the total cost while avoiding obstacles and risky areas.

Environment Type

- **Partially Observable:** The robot cannot see the entire building.
- **Dynamic:** Conditions may change during the rescue operation.
- **Sequential:** Every action affects the next decision.
- **Single Agent:** Only one rescue robot performs the task.

(d) Justification

Uniform Cost Search is the most suitable algorithm because it always selects the path with the **lowest cumulative cost**, considering both movement cost and risky zones. An **Online Search Agent** is appropriate because the robot has limited knowledge of the environment and must continuously update its decisions based on newly observed information.

Final Answer

- **Search Strategy:** Uniform Cost Search (UCS) with an Online Search Agent
- **Agent Type:** Intelligent Agent
- **Environment:** Partially Observable, Dynamic, Sequential, Single Agent
- **Objective:** Reach the survivor safely while minimizing the total cost and avoiding obstacles and risky zones.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided